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PII: S1385-8947(18)30298-5
DOI: <https://doi.org/10.1016/j.cej.2018.02.085>
Reference: CEJ 18564

To appear in: *Chemical Engineering Journal*

Received Date: 28 December 2017
Revised Date: 31 January 2018
Accepted Date: 19 February 2018



Please cite this article as: D. Wang, Y. Wang, H. Liu, W. Xu, L. Xu, Unusual carbon nanomesh constructed by interconnected carbon nanocages for ionic liquid-based supercapacitor with superior rate capability, *Chemical Engineering Journal* (2018), doi: <https://doi.org/10.1016/j.cej.2018.02.085>

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Unusual carbon nanomesh constructed by interconnected carbon nanocages for ionic liquid-based supercapacitor with superior rate capability

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The exploration of various porous nanocarbon materials with an extraordinary electrochemical capacitive performance in ionic liquid (IL) electrolytes is urgently needed for development of the next-generation energy storage devices. Significant improvements in energy density have been achieved, however, supercapacitors with IL electrolytes usually suffer from issues related to low power density and large IR drop. Herein, we report an unprecedented carbon nanomesh that is made up of interconnected densely-packed carbon nanocages with ultrathin partially graphitic shells. The large specific surface area and well-developed mesopores combined with its unique structural features, make it a very competitive material for high-performance supercapacitor in IL electrolyte. Notably, this unique nanosheets morphology with lots of in-plane nanopores on their surface can provide fast cross-plane ion diffusion pathways, and meanwhile, the carbon nanocages with partially graphitic shells ensure the rapid electron transfer. The supercapacitor assembled with the optimized sample can deliver a high specific capacitance (up to 194 F/g at 1 A/g), superior rate capability (68% capacitance retention at 70 A/g), low IR drop (0.685 V at 70 A/g). Moreover, benefiting from the wide working voltage and high-rate capability, the energy densities can maintain at 56.1 Wh/kg even at an ultrahigh power density of 61.250 kW/kg. Strikingly, the as-assembled supercapacitor can successfully power a light-emitting diodes (LED, 2.2 V) module with 53 red LED lights demonstrating its great potential for practical applications in energy storage devices.

Keywords: Carbon nanomesh; Carbon nanocages; Ionic liquid; Supercapacitors; Rate

capability

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1. Introduction

Developing sustainable and renewable electrochemical energy storage materials is of significant importance to meet the ever-increasing demand for high-performance multifunctional energy storage and conversion devices [1,2]. Among the various available electrochemical energy storage systems, supercapacitors has gained tremendous attention for energy harvesting in view of their fantastic properties such as ultrahigh power density, ultralong cycle life and low maintenance costs [3-7]. Generally, activated carbons (AC) have been extensively evaluated as electrode materials for commercial supercapacitors given their wide availability, large specific surface area, moderate conductivities, favorable physicochemistry stability, and tunable surface properties [8-10]. However, the extremely sluggish ion transport that even inaccessible for electrolyte ions at high rate lead to poor diffusion kinetics, and therefore unsatisfied capacitance and rate capability.

Recent years have witnessed an explosive growing in exploring of various carbonaceous materials ranging from zero-dimensional (0D), one-dimensional (1D), and two-dimensional (2D), to three-dimensional carbons in order to further improve the capacitive performance particularly at high rates [11-20]. The substantially improved performance can be attributed to their unique morphologies and textural features that can provide abundant active sites for charge storage to obtain large capacitance and fast ions diffusion channels to ensure good rate performance. As a new type of 2D carbon materials, carbon nanomesh (CNM) began to receive a lot of attention since the discovery of graphene nanomesh (also known as holey graphene)

[21, 22]. Fundamentally, carbon nanomesh has a very competitive electrochemical performance than that of the others 2D carbons due to the substantially improved cross-plane mass transfer while the severely agglomeration and stacking can be significantly alleviated as compared with graphene nanomesh. Nevertheless, the controllable synthesis of carbon nanomesh with tunable pores is a very challenging task because of the requirement for simultaneously formation of nanosheets and in-plane pores on their surface. In comparison with the studies on the graphene nanomesh, efforts towards the synthesis and electrochemical performance of carbon nanomesh have lagged far behind. For example, the Qiu and Lei research group reported a facile chemical vapor deposition method to prepare carbon nanomesh with MgAl-layered double oxides as sacrificial template and ferrocene as carbon precursor [23]. As supercapacitor electrode, the CNM exhibits an enhanced capacitive performance. Recently, our research group developed a novel explosion-assisted activation process employing KNO_3 as both an explosive and activating reagent for the synthesis of CNM, and the results revealing that the CNM displays excellent electrochemical performance for supercapacitors in organic electrolyte [24]. Therefore, developing various routes toward CNM and further illuminating its enhanced electrochemical performance is great of interesting but still remain a tremendous challenge.

Carbon nanocages (CNCs) with a thin graphitic wall and a hollow interior have received considerable research attention because of their unique geometrical structure and extensive applications in many fields, such as adsorbents [25], catalyst support

[26], lithium-ion batteries [27], and supercapacitors [28]. These promise has triggered intense research and development efforts by ever-increasing research groups seeking to exploit a wide variety of strategies for preparing CNCs and studying its enhanced performance. Considering its large specific surface area, good electrical conductivity as well as excellent physicochemistry stability, the CNCs usually shows improved electrochemical performance [29]. For instance, Hu and coworkers developed an MgO template method for the producing of CNCs with high specific surface areas, and the product presents remarkable performance as the electrode materials of supercapacitor [28,30]. Ma et al. reported a facile one-step route by graphitizing 1-hexadecylamine with CoCl_2 for the synthesis of CNCs, and it exhibits excellent electrochemical performance when used as anode material for lithium-ion batteries [31]. A majority of the previous methods, however, usually involves complicated hard template-assisted process and the sizes of CNCs are highly dependent on the size of template used, which leads to poor size and uniformity control, especially for small CNCs [32]. Thus, it is utmost importance but a great challenge to develop a facile route to CNCs in an attempt to further utilize their unique properties. Furthermore, assembling of these nanocages into organized structures would be shed a new insight for fully discovering their unprecedented properties because of the assemblies usually exhibit newer and/or better properties different from the individual counterparts [33,34]. However, little success has been achieved for the synthesis of carbon nanostructures employing nanocages as the building blocks owing to the tremendous complexity associated with the adjusting and assembling of the nanocarbons.

Supercapacitors with room temperature ionic liquid (IL) electrolytes hold great promise for the development of next-generation high-performance energy storage devices in view of their wide stable work voltage along with the ultrahigh energy density [35]. However, ILs has large ionic size usually exhibiting high viscosity and low conductivity thus restrict the diffusion of ions into the narrow micropores, and thereby the unsatisfactory electrochemical performance especially at high rate as compared with those using conventional aqueous electrolytes [36,37]. Therefore, it is of great importance to exploit new nanostructured porous carbons with optimized pore size to accommodate the large IL ion and large accessible specific surface area for charge storage, and meanwhile, the partial graphitized domains to ensure rapid charge migration. Nevertheless, this still remains a highly sophisticated challenge.

Herein we firstly demonstrated an unprecedented carbon nanomesh constructed by small carbon nanocages (designated as CMCNCs) through the direct carbonization of the intermediate products obtained from $\text{Ni}(\text{NO}_3)_2$ and sucrose. The carbon nanomesh has lots of in-plane nanopores that can provide fast ion diffusion pathways across the nanosheets while the smaller nanocages with ultrathin and partially graphitic shells can substantially accelerate the electron transfer that will boost the electrochemical performance distinctly. As a proof of concept, this novel CMCNCs nanostructure exhibits tremendously improved capacitive performance in neat IL electrolyte as compared with commercial activated carbon and shows large specific capacitance (up to 194 F/g at 1 A/g), superhigh rate capability (68% capacitance retention at 70 A/g), tiny IR drop (0.685 V at 70 A/g) at large current densities and excellent cycling

performance (more than 90% capacitance retention after 10000 cycles).

2. Experimental Section

2.1. Materials preparation

All chemicals used were analytical grade (purchased from Aladdin Chemical Reagent Co., Ltd. Shanghai, China) and used as received without any further purification.

CMCNCs were synthesized by direct carbonization and graphitization of the intermediate products obtained from $\text{Ni}(\text{NO}_3)_2$ and sucrose where $\text{Ni}(\text{NO}_3)_2$ serves as the oxidizing agent and catalyst for graphitization. In a typical process, 7 g of nickel nitrate hexahydrate ($\text{Ni}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$) and 10 g of sucrose are dissolved in 15 mL of deionized water to form a homogeneous solution (Fig. S1a). Then, the solution was transferred into a specially designed quartz bowl and put into the preheat oven at 200 °C. After twenty minutes, the earthy yellow product with a large volume expansion was obtained (Fig. S1b). Subsequently, this earthy yellow colored intermediate product was crushed in an agate mortar and pyrolysis in a horizontal tubular furnace equipped with a corundum tube (Kejing Group, China) at 600 °C for 2 h with a ramping rate of 2 °C/min under an argon flow. After cooling down to room temperature naturally, the black products were treated with concentrated HCl (12 M) to remove the nickel residues, and washed with abundant deionized water until pH value of the filtrate became neutral. Finally, the products were dried in an oven at 80 °C for 12 h. For simplicity, the as-obtained samples were described as CMCNC-X in subsequent discussions, where X is the mass of $\text{Ni}(\text{NO}_3)_2$ (g) used (X = 3, 5, 7, and

9 is corresponding to the mass of $\text{Ni}(\text{NO}_3)_2$ is 3, 5, 7 and 9 g, respectively.).

2.2. Structure characterization

The morphologies and microstructures of the products were characterized by scanning electron microscope (SEM, Zeiss Supra55, Germany) and transmission electron microscope (TEM, JEOL JEM-2100F, Japan). Powder X-ray diffraction (XRD) measurements were carried out on a Shimadzu X-ray 6000 diffractometer with Cu K α radiation (wavelength = 1.54 Å) between 5° and 80° at a step width of 0.02 degree. Room-temperature Raman spectra measurements were performed on an InVia Raman microscope (Renishaw, England) with a laser wavelength of 532 nm. The surface elemental compositions and chemical states were analyzed by the X-ray photoelectron spectroscopy (XPS, Thermo Fisher Scientific, USA) with an Al K α X-ray source. The binding energy was corrected by referring to the C 1s peak at 284.8 eV. The specific surface area, pore volume and pore size distribution of the samples were determined by N₂ adsorption–desorption at 77 K (ASAP 2020, Micromeritics, USA) after being degassed at 150 °C overnight. The specific surface area of the samples was obtained from Brunauer-Emmett-Teller (BET) theory and the pore size distribution was estimated by adsorption branch from the Barrett-Joyner-Halenda (BJH) method.

2.3. Electrochemical measurements

Working electrodes were prepared by mixing the as-obtained CMCNCs materials with carbon black (super P) and poly (tetrafluoroethylene) binder at a mass ratio of 90:5:5 in ethanol to form a homogeneous slurry. Afterwards, the mixture was rolled

into thin carbon sheet with a roll squeezer (Kejing Group, China), then punched into circular electrodes with 14 mm in diameter. After drying in vacuum at 80 °C for 12 h, the electrodes were pressed onto aluminum foil at a pressure of 10 MPa to ensure a good mechanical adhesion between the active materials and the current collectors. The mass loading of active material on each working electrodes was about 3-4 mg. The test CR2025 coin-type cells were assembled in an argon-filled glove box (Kejing Group, China, both O₂ and H₂O contents lower than 0.1 ppm) with two identical electrodes separated by a separator (NKK, TF4030). The neat ionic liquid 1-ethyl-3-methylimidazolium tetrafluoroborate (EMIMBF₄, 98%) was used as the electrolyte. AC with a very high specific surface area up to 1822 m²/g for commercial supercapacitors was also tested in the same experimental conditions for comparison.

Electrochemical performances for the as-synthesized products were carried out on a CHI 660E electrochemical workstation (Shanghai Chenhua Co., China) at ambient temperature. Cyclic voltammetry (CV) studies were performed with the scan rates from 10-1000 mV/s in the potential range of 0-3.5 V. Galvanostatic charge-discharge measurements were conducted at current densities ranged between 1 and 70 A/g based on the mass of active material in a single electrode. Electrochemical impedance spectroscopy (EIS) measurements were carried out in the frequency range of 10⁵-10⁻² Hz with a 5 mV AC amplitude. The cyclic stability testing was performed on a LAND CT2001A instrument (Wuhan Land Co., China) at 5 A/g within 0-3.5 V.

The specific capacitance of the electrode materials was calculated according to the galvanostatic discharge process based on the equation $C_s = 2(I\Delta t/3.5m)$, where, C_s is

the specific capacitance (F/g), I is the discharge current (mA), Δt is the discharge time (s), m is the mass of active material in a single electrode (mg), and 3.5 is operation potential window (3.5 V), respectively. The energy density (E , Wh/kg) and the power density (P , W/kg) of the symmetrical supercapacitors were calculated according to $E = Cs3.5^2/8$ and $P = E/\Delta t$, respectively.

3. Results and discussion

3.1. Formation and structural characteristics of the CMCNCs

Fig. 1 shows the schematic illustration for the fabrication process of CMCNCs from $\text{Ni}(\text{NO}_3)_2$ and sucrose. Firstly, nickel nitrate hexahydrate ($\text{Ni}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$) and sucrose are dissolved in small amount of water to form a homogeneous solution, and the water evaporated quickly at 200 °C. Secondly, Above 160 °C, sucrose began to melt and dehydration while the $\text{Ni}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ start to decompose that induce violent and complex reaction together with released a large amounts of gas (e.g. NO_2 , NO , O_2 , CO_2 , and CO . Detailed chemical reactions and calculations based on thermodynamic theory are listed in table S1 in supplementary data) give rise to earthy yellow intermediate product with a large volume expansion. It is worth to be noted that this process is similar to the chemical blowing process which is very helpful for the formation of sheet-like structures [38,39]. Thirdly, during pyrolysis at high temperature nickel nanoparticles formed base on the reaction: $\text{NiO} + \text{C} = \text{Ni} + \text{CO}$ (The thermodynamic parameter calculation is listed in supplementary data). Importantly, partial amorphous carbon formed at this stage gradually to dissolve in the nickel particles, and then precipitated as graphitic nanostructures around the nickel

nanoparticles leading to the formation of thin-walled graphitic nanoshells according to the dissolution–precipitation mechanism (Fig. S2) [40]. Finally, the nickel nanoparticles were removed by react with hydrochloric acid, and left the CMCNCs nanostructures.

Fig. 2, Fig. S3-S7 show the SEM images of the as-prepared products. Among them, the CMCNC-7 sample exhibits a sheet-like morphology with many evenly distributed holes, and thickness of about 100 nm (Fig. 2a, 2b and Fig. S3a). A closer SEM examination further revealed that the nanosheets has a rough surface and the pore size of the nanoholes about 30-50 nm (Fig. S3b). TEM image reveals that the sheet-like structure is actually constructed by numerous densely-packed carbon nanocages, and there are also some nanoholes with pore size about 30-50 nm on its surface (Fig. S4, Fig. 2c). Notably, high resolution TEM analysis indicated that the carbon nanocages has ultrathin partial graphitic shells with mostly has a few layers graphene (Fig. 2d). This unique morphology contain lots of in-plane nanopores that could be helpful for fast transport of electrolyte ions into the inner pores, and the densely-packed smaller nanocages with ultrathin partial graphitic shells can offer a continuous conductive network for rapid electrons transfer, which is expected to boost the electrochemical performances remarkably. Moreover, we found that the content of the $\text{Ni}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ play a vital role for the formation of CMCNCs. Well-defined CNM cannot be obtained at a low $\text{Ni}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ content (e.g. 3 g) (Fig. S5), and start to formation as the $\text{Ni}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ content increased to 5 g (Fig. S6). If the $\text{Ni}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ content further increased to 9 g, the product mainly composed carbon nanofibers (Fig. S7).

However, the earthy yellow colored intermediate product burn vigorously at 200 °C in the oven and only Ni-NiO nanocomposite was obtained if large $\text{Ni}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ content was employed (e.g. 10 g) because of the Ni species holds high catalytic activities (Fig. S8) [41]. It is worth to be noted that the Ni-NiO nanocomposite also shows a nanomesh structure, which can served as the template for the formation of carbon nanomesh (Fig. S9).

The phase structures of the as-synthesized products were investigated by XRD measurement. As shown in Fig. 3a, two broad diffraction peaks at around 23.4° and 43.9°, which can be assigned to the diffractions of (002) and (100) planes of graphite, respectively. The broad diffraction peak at 23.4° slightly increased as the content of $\text{Ni}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ increased can be attribute to the enhancement of the catalytic activity. Moreover, the great increased diffraction intensity in the low angle region ($< 10^\circ$) indicates the presence of pores in large quantities, which is consistent with the SEM and TEM observation. Raman spectroscopy was carried out to further evaluate the structures of these carbon samples. As displayed in Fig. 3b, all the samples exhibit two strong peak located at 1350 cm^{-1} and 1595 cm^{-1} , which is correspond to the D and G bands of carbonaceous materials, respectively [42]. Generally, the G band is related to the ordered graphitic structure, while the D band is associated with the disordered and defects structures in in carbon materials [43]. The integral intensity ratio of D band and G band (I_D/I_G) is usually used to estimate the level of the graphitization degree of carbon materials. The I_D/I_G ratios are 0.96 for CMCNC-3, 0.95 for CMCNC-5, 0.93 for CMCNC-7, and 0.90 for CMCNC-9. The slightly decreased I_D/I_G

ratios as the content of $\text{Ni}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ increased, confirming the percentage of ordered carbons increased, which is good consistent with the XRD results. The surface chemical composition and states of the CMCNC-7 were determined by XPS analysis and is presented in Fig. 3c. The XPS survey spectra show two dominant peaks centered at 284.8 and 532.3 eV corresponding to C1s and O1s, respectively. The corresponding concentrations of the C and O elements are 94.78 at.% and 5.22 at.%. The low heteroatom content can not only improve the conductivity and the stability but also lower the self-discharge of the electrodes, eventually, effectively improves the electrochemical performance of the device [44,45]. The deconvolution of C1s spectra include sp^2 C at 284.8 eV, sp^3 C at 285.4 eV, C-O at 286.5 eV and C=O/COOH at 289.8 eV (Fig. 3d).

Textural characteristics of as-obtained porous carbons were investigated by N_2 adsorption/desorption measurements. The isotherms and the corresponding pore size distributions (PSDs) estimated by means of the BJH method are shown respectively in Fig. 4a and 4b. It can be seen that all the samples shows a distinct type IV isotherm with evident hysteresis loops according to IUPAC classification [46]. The large hysteresis loops are generally associated with capillary condensation in mesopores, which is the typical features of mesoporous materials [47]. Compared with other samples, CMCNC-7 has a relatively large adsorption at low pressures suggests the existence of a number of narrow nanopores less than 2 nm than that of the others (Fig. 4a). Moreover, the accretion uptake at the high relative pressure indicating the presence of abundant larger pores due to the hollow structure of the carbon nanocages,

which is consistent with TEM studies. Fig. 4b shows that all porous carbons display a broad pore size distribution (calculated by BJH method) and the majority of the nanopores in the range of 2-5 nm while the proportion of small nanopores (between 1-2 nm) slightly increased from CMCNC-3 to CMCNC-7 then decreased for CMCNC-9. The mesopores of CMCNC-3 peaked at 2.4 and 4.0 nm, while the mesopores of CMCNC-5 and CMCNC-7 is smaller (2.6 nm and 3.3 nm for CMCNC-5, 2.2 nm for CMCNC-7), and widen again for CMCNC-9 (2.3 and 3.5 nm). The obvious variance in PSDs of the samples should be attributed to the different morphologies. The specific surface area of CMCNCs are calculated to be 774, 983, 1198, and 433 m²/g, respectively, mesopores account for most of specific surface area (see Fig. 4c and Table 1). The increased specific surface area at higher Ni(NO₃)₂·6H₂O content resulting from the more developed pores while the significant loss of the specific surface area on account of the formation of nanofibers as the Ni(NO₃)₂·6H₂O content further increased to 9 g. The large specific surface area with well-developed mesopores structure and fewer micropores make CMCNC-7 a very promising electrode material for high-performance ionic liquid based supercapacitor.

3.2. Electrochemical performance of the CMCNCs

The electrochemical performance of as-obtained CMCNCs were evaluated in a symmetric two-electrode system with the neat ionic liquid EMIMBF₄ as the electrolyte. AC for commercial supercapacitors was also tested in the same experimental conditions for reference. Fig. 5a shows the CV curves of CMCNCs and AC with the scan rate of 100 mV/s over a wide voltage window from 0 to 3.5 V.

Evidently, the CMCNC-7 exhibit the best electrochemical performance owing to the better rectangular-like shape and larger enclosed area of the CV curves, while the specific capacitance decreased in the in the sequence of CMCNC-7 > CMCNC-5 > CMCNC-3 > AC. The absence of redox peaks suggesting that the capacitance mainly resulting from electrical double-layers (EDLCs) due to the low heteroatom content. Moreover, the CV curves of CMCNC-5 and CMCNC-7 samples can remain the quasi rectangular profile while the CMCNC-3 start to distortion especially for AC at the scan rates of 100 mV/s. Fig. 5b and Fig. S10 shows the CV curves for CMCNC-7 electrode at the different scan rates range from 10 to 1000 mV/s. Obviously, the CV curves show a fairly rectangular shape at low scan rates (10-100 mV/s) indicating the effective for the formation of electric double layers at the electrode-electrolyte interface. Fascinatingly, the rectangular shape was retained even at an ultrahigh scan rate (1 V/s) suggesting that the excellent capacitive performance and super rate capability, which can be attributed to the unique morphology with many in-plan nanopores that facilitating for the rapid transfer of ions and electron. For CMCNC-3 and CMCNC-5, the rectangular shape can only be maintained at low scan rates (10-100 mV/s), and a severe distortion can be observed with the further increase in the scan rate, this is especially the case for CMCNC-3 (Fig. S11 and Fig. S12). AC based electrode, however, displayed significantly distortion in the CV curves even at a low scan rate (10-100 mV/s, Fig. S13) mainly due to the narrow micropores significantly slow down the ion diffusion ability as well as the large number of ultra micropores that are inaccessible to electrolyte ions (Fig. S14) [48]. All of the

CMCNCs have better rate capability performance compared with AC supporting that the porous carbons containing a large amount of mesopores (2-50 nm) is the basic requirement for efficient transport of IL electrolyte ions to ensure the power capability. Moreover, the rate capability can be further improved by introducing additionally ion diffusion channel (such as, in-plane nanopores) and partial graphitized domains to guarantee the rapid ions and electron transfer. The galvanostatic charge-discharge profiles of the CMCNCs at various current densities are shown in Fig. 5c and 5d. Apparently, the CMCNC-7 electrode displays the largest specific capacitance compared with other CMCNCs electrodes as well as AC, which is good agreement with the CV results. As shown in Fig. 5c, CMCNC-7 based electrode exhibits highly linear symmetric triangular-shaped curves, without obvious IR drop at various current densities vary from 1 to 10 A/g, revealing its excellent electrochemical reversibility and tiny internal resistance. Notably, the charge-discharge curves are still maintain symmetric with low IR drops as the current densities further increased, the relative small IR drop of 0.68 V even at ultrahigh currents density of 70 A/g (Fig. 5d), further demonstrating that the CMCNC-7 based device has a low equivalent series resistance (ESR). The relationship between the IR drop and current densities are further displayed in Fig. 5e. The IR drop is calculated to be 0.21, 0.20 and 0.10 V for CMCNC-3, CMCNC-5 and CMCNC-7 electrodes at 10 A/g, respectively. When the current density was increased to 30 A/g, CMCNC-7 electrode has a lower IR drop of 0.286 V compared with CMCNC-5 (0.474 V) where the IR drop as large as 0.61 V for CMCNC-3 electrode. Remarkably, the IR drop only slightly increased to 0.685 V as

the current density was increased to 70 A/g (seventy-fold increased), demonstrating that the CMCNC-7 can afford large energy density at ultrahigh output power. For AC electrode, however, a large IR drop up to 1.328 V even at a low current density of 5 A/g. According to the linear relationship ($R^2 > 0.99$) between current densities and the IR drop values, the internal resistance of CMCNC-7 electrode is much lower than that of the CMCNC-5 and CMCNC-3 electrode, while the internal resistance in the AC electrode is considerable large. The small IR drop for CMCNC-7 electrode at variety of current densities can be attributed to the synergistic effect of the densely-packed carbon nanocages with partially graphitic shells and the nanomesh structure that can provide rapid transport pathways for electrons and ions simultaneously, thus can afford a large power delivery without sacrificing the original high energy densities of the supercapacitors in IL based electrolyte. The specific capacitance at various current densities was calculated and the results are shown in Fig. 5f. The specific capacitance of 100, 167, 194 and 77 F/g are obtained for CMCNC-3, CMCNC-5, CMCNC-7 and AC at the current density of 1 A/g, respectively. Obviously, the CMCNC-7 has the highest specific capacitance among all of the electrodes (Fig. S15). Moreover, it also displays excellent specific capacitance retention of 84% and 78.9% at 30 A/ and 40 A/g, much higher than that of CMCNC-3 (50% at 30 A/g) and CMCNC-5 (69.9% at 40 A/g). Strikingly, the specific capacitance still maintained 132 F/g even at an ultrahigh current density of 70 A/g with a remarkable specific capacitance retention of 68%, which is much better than those of other advanced porous carbons in various kinds of electrolytes reported in previous works during the last three years (Table

S2). Furthermore, the CMCNC-7 based supercapacitor also exhibited good cycling stability during the continuous galvanostatic charge-discharge cycles at a moderate current density of 5 A/g (Fig. 5g). The specific capacitance still remained 90.6% of its initial specific capacitance after 10000 cycles, demonstrating its excellent electrochemical stability. From the insert in Fig. 5g, the linear symmetric triangular-shaped curves changed a little after cycles although the IR drop increased slightly.

Electrochemical impedance spectroscopy (EIS) was performed to further evaluate the ion transport kinetics and electron transfer behavior within the electrodes. Fig. 6a shows the Nyquist plots of the CMCNCs, and the inset is the magnified high-frequency region. Generally, the Nyquist plot mainly composed of three parts: an intercept in the high frequency region, a semicircle in the middle frequency region, and an inclined line in the low frequency region. Normally, the intercept in the high frequency region corresponds to internal resistance (R_s), and the semicircle in the middle frequency region is attributed to charge-transfer resistance (R_{ct}) at the electrode/electrolyte interface while the inclined line vertical line in the low frequency region can be assigned to Warburg impedance (R_w) [49-52]. The difference in the resultant internal resistance and charge-transfer resistance are small but variation in Warburg impedance association with the diffusion process suggesting that the capacitive performance determined by ions migration. This result suggesting that the ion diffusion resistance played a very dominant role for the carbons with graphitic or partially graphitic structure. Herein, we defined the ESR is the sum of R_s , R_{ct} and R_w ,

which is one of the most critical parameter that determining the electrochemical performance of the EDLCs. The ESR are calculated to be 2.85, 2.31, and 1.92 Ω for CMCNC-3, CMCNC-5, and CMCNC-7 based supercapacitors, respectively. The lowest ESR for CMCNC-7 based electrode supporting again that the CMCNC-7 displays excellent capacitive performance in IL electrolyte, which is in good agreement with the CV and galvanostatic charge-discharge results. The Bode phase plot (Fig. 6b) show that the phase angle of CMCNC-7 is about 86.1° at 10⁻² Hz, which approach to 90° for an ideal supercapacitor behavior [53]. The characteristic frequency (f_0) at a phase angle of 45° is 0.41 Hz for the CMCNC-7 and the corresponding time constant t_0 ($t_0 = 1/f_0$) is calculated to be about 2.43 s, nearly half of the value for CMCNC-5 electrode (0.21 Hz, 4.76 s), further supporting that the in-plane nanopores can afford an efficient and rapid transport pathways for ion diffusion. All these results further demonstrated the excellent electron and ion-transport behavior of CMCNC-7 based supercapacitor.

Fig. 7a shows the relationship between energy density and power density, also called Ragone plot, which is very useful to evaluate the overall performance of the energy storage systems. It is worth noting that the assembled CMCNC-7 supercapacitor exhibits a higher energy density of 82.4 Wh/kg at a power density of 0.875 kW/kg, and still preserved at 56.1 Wh/kg at an ultrahigh power density of 61.25 kW/kg, which are comparable or even superior to most of those recently reported carbon-based supercapacitors in IL electrolyte, such as hierarchically porous sheet-like nanocarbons (51.7 Wh/kg at 35 kW/kg) [54], F, N co-doped carbon

nanofibers (8.07 Wh/kg at 0.248 kW/kg)[55], reduced graphene oxide (131.8 Wh/kg at 0.892 kW/kg)[56], lignin derived hierarchical porous carbon (22.9 Wh/kg at 25.4 kW/kg) [57], polyaniline derived hierarchical porous carbon (10 Wh/kg at 10 kW/kg) [58], polysiloxane derived porous carbon (21 Wh/kg at 30 kW/kg) [37], N-doped carbon nanosheets framework (60.4 Wh/kg at 1.75 kW/kg) [59], and so on [36,60-71]. Amazingly, benefiting from the integrated high energy and power properties, single symmetrical supercapacitor device based on CMCNC-7 electrode can successfully power a light-emitting diodes (LED, 2.2 V) module with 53 red LED lights for more than 2 min after charging at 5 A/g for 30 s (Fig. 7b and Video S1). All of these results evidently support that CMCNC-7 obtained from such a versatile process have great potential for high-performance supercapacitor application in IL electrolyte.

4. Conclusions

To sum up, novel porous carbon nanomesh constructed by interconnected densely-packed carbon nanocages are prepared to be efficient electrode materials for IL based supercapacitors through a facile method by employing sustainable sucrose as precursor and $\text{Ni}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ as oxidizing agent and catalyst for graphitization. The as-obtained CMCNC-7 possess large surface and very high of mesoporous content that make it suitable for IL based supercapacitors. Elegantly, benefiting from multi synergetic effect of in-plan nanopores and carbon nanocages with partially graphitic shells, the CMCNC-7 material delivered extraordinary capacitive performance in IL electrolyte, such as large specific capacitance up to 194 F/g, superior rate capability, remarkable energy density of 20.2 Wh/kg, and excellent cycle stability up to 10000

cycles, which puts the CMCNC-7 among best capacitive performance in the reported of the state-of-the-art porous carbon-based supercapacitors in IL electrolyte. Most importantly, our work prove that the CMCNCs with integrated features of nanomesh and nanocages is a new pathway to design and controllable synthesis of porous carbon nanoarchitectures for high-rate supercapacitors in IL electrolyte.

Acknowledgements

The authors are grateful to the financial supports from the National Natural Science Foundation of China (Grant no. 51762001) and the Key Scientific Research Projects at North Minzu University (Grant no. 2017KJ06).

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version.

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Figures

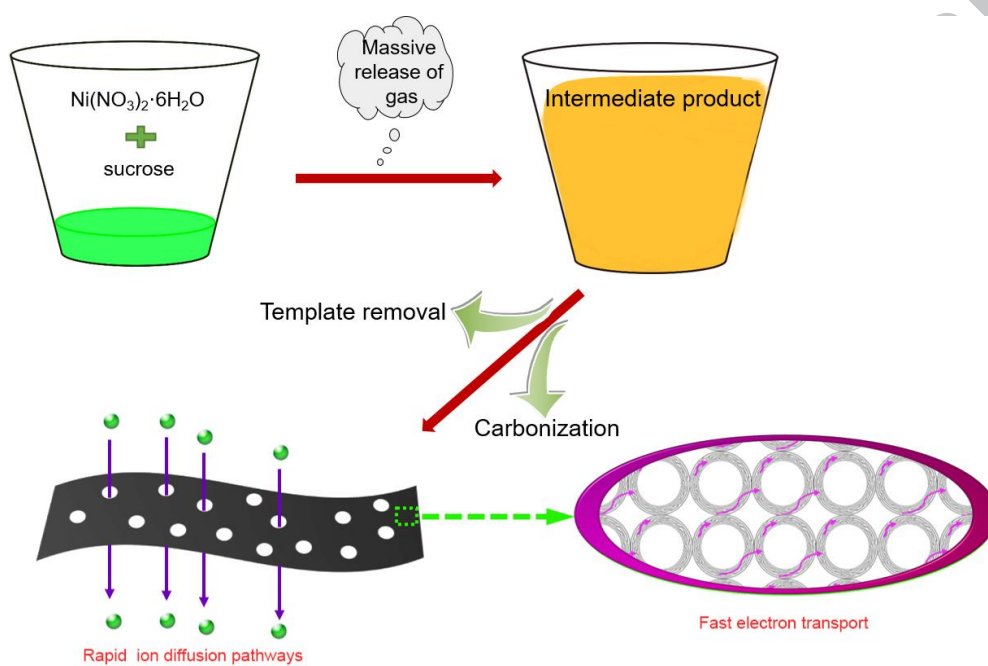


Fig. 1. Schematic illustration of the fabrication and ions-electrons transportation through CMCNCs.

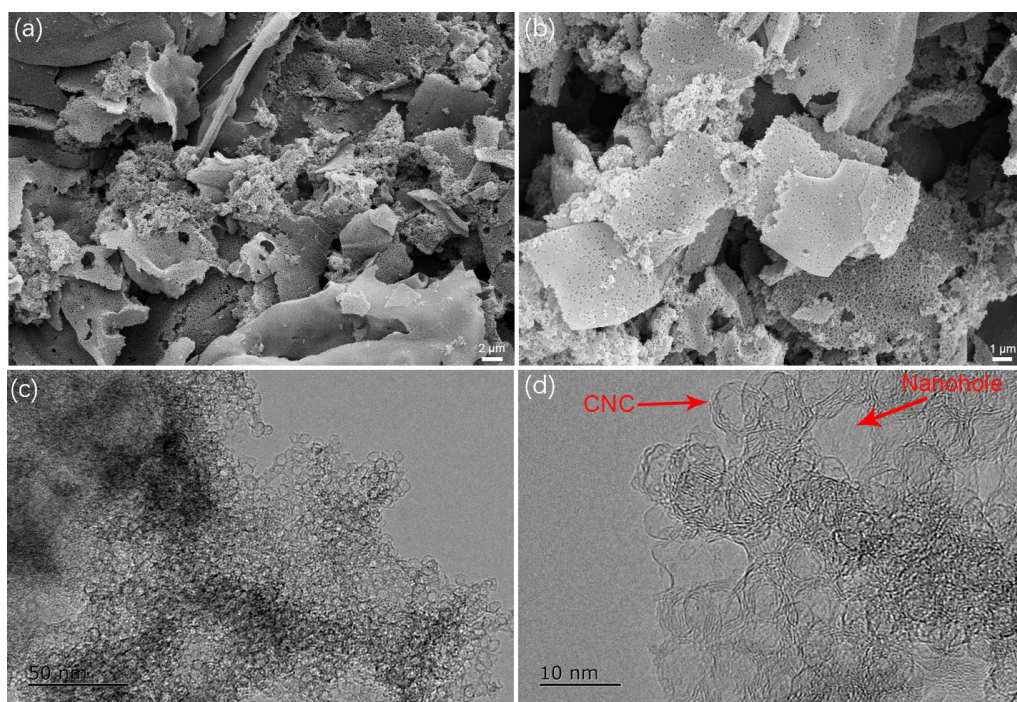


Fig. 2. Morphological characterization of the CMCNC-7. (a), (b) SEM images, (c) TEM images, and (d) high resolution TEM image.

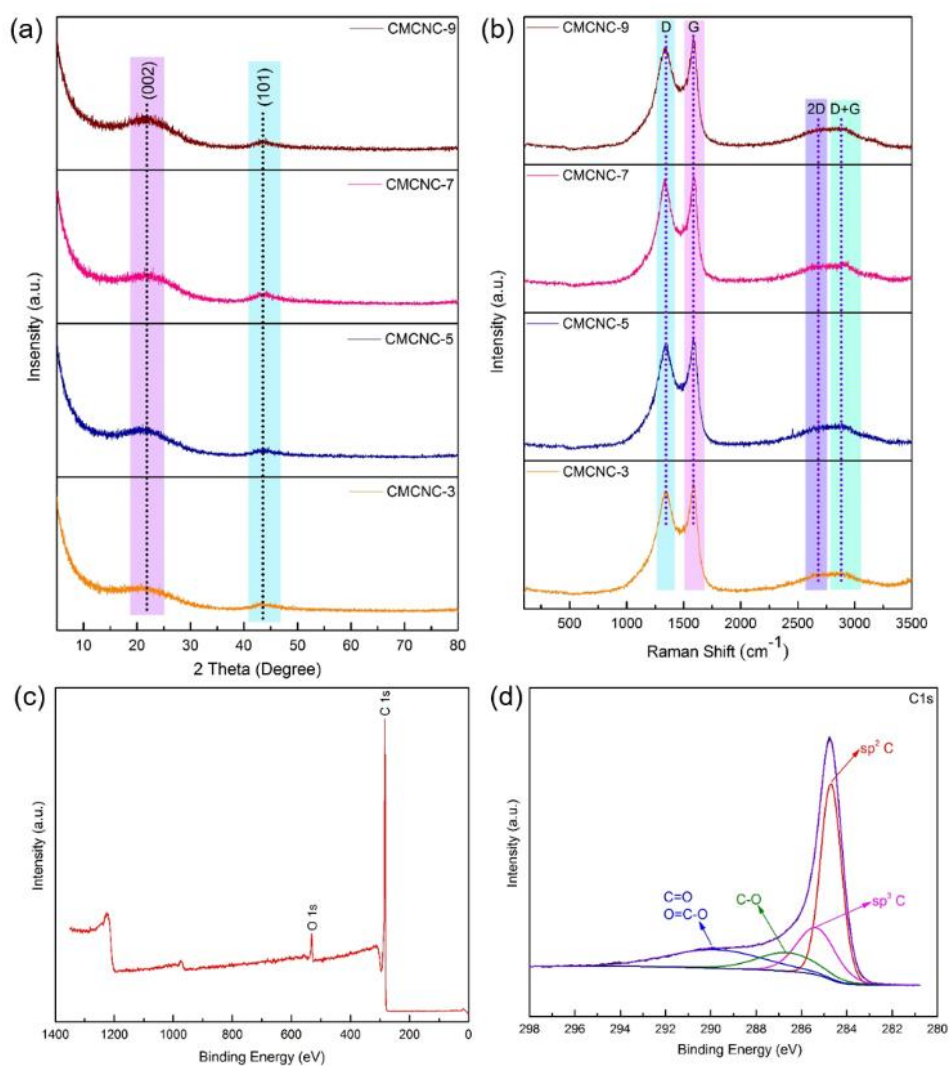


Fig. 3. XRD patterns (a) and Raman spectra (b) of the CMCNCs, (c) XPS spectra and high-resolution C 1s spectra of the CMCNC-7.

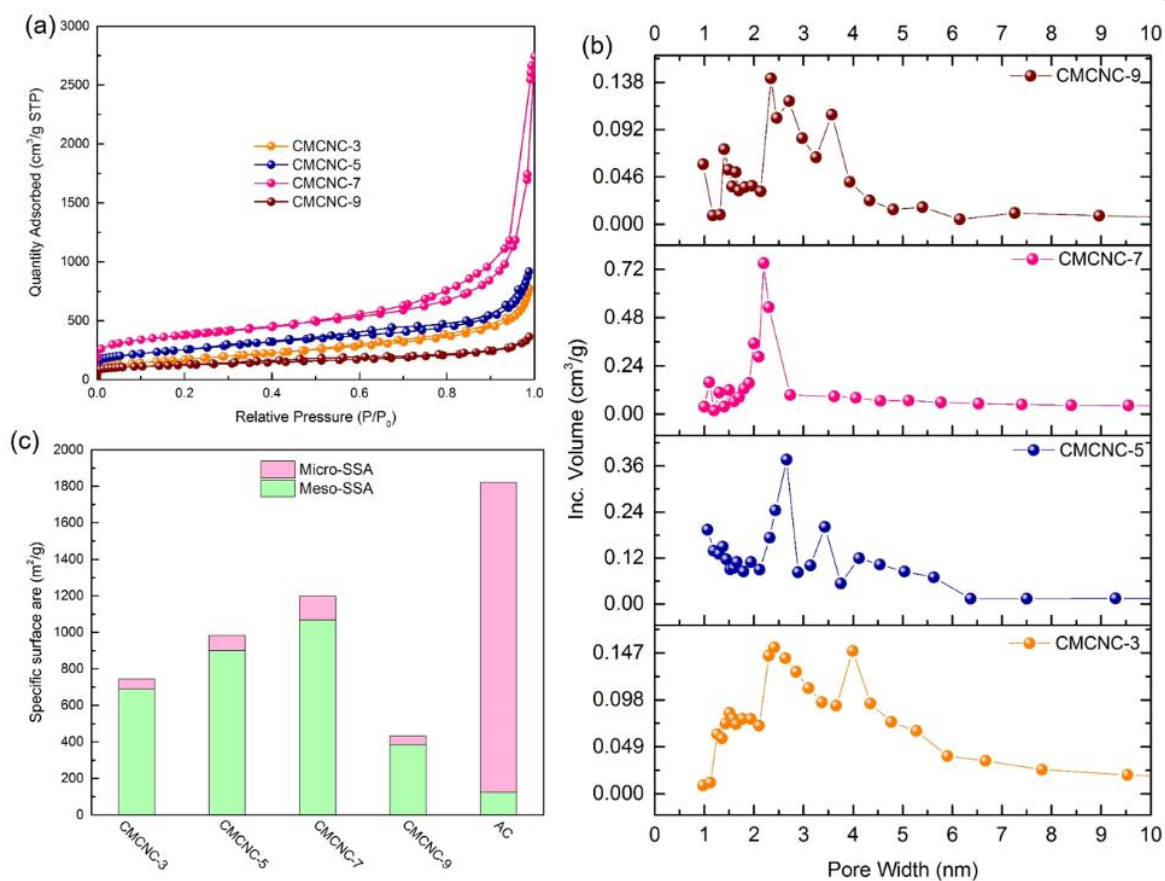


Fig. 4. Textural characterization of the CMCNCs. (a) N₂ adsorption-desorption isotherms, (b) the corresponding pore size distribution profiles based on BJH method, (c) relative content of the mesopores and micropores surface areas.

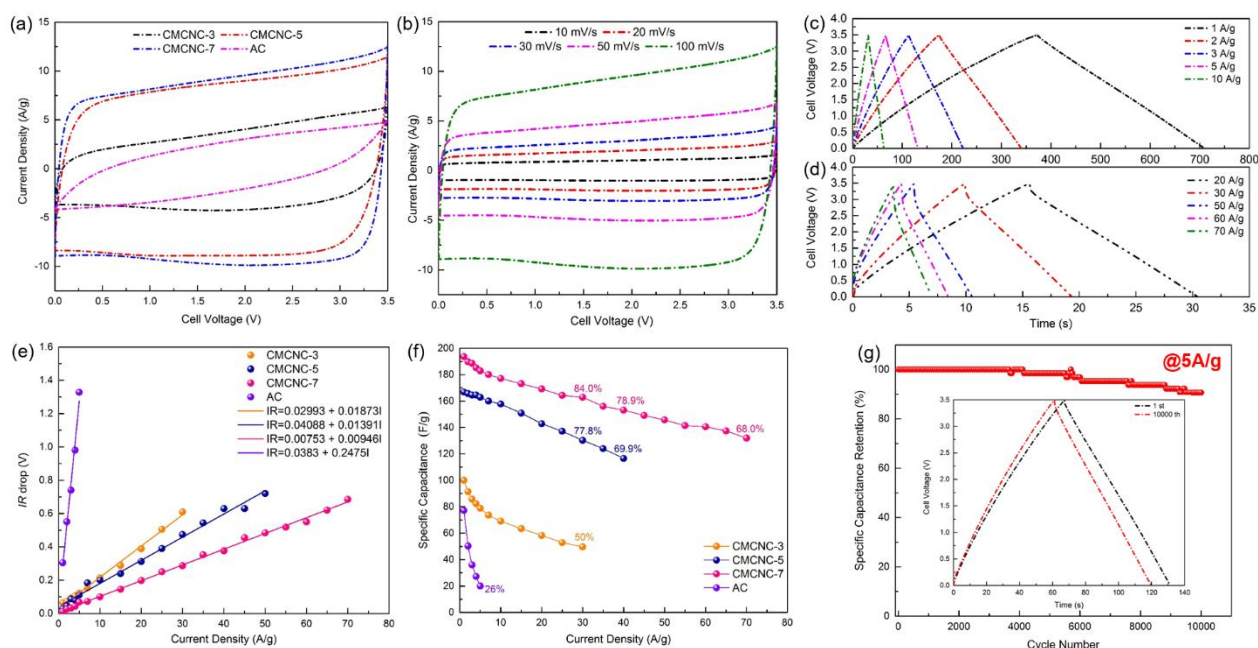


Fig. 5. Electrochemical characterizations of the CMCNCs and AC. (a) CV curves of the CMCNCs and AC at a scan rate of 100 mV/s, (b) CV curves of the CMCNC-7 at the scan rates ranging from 10 to 100 mV/s, (c) and (d) galvanostatic charge-discharge curves at various current densities, (e) IR drop at different current densities, (f) The specific capacitances at various current densities, and (g) Cycling stability of the CMCNC-7 based electrode at 5 A/g for 10000 cycles. Inset is the galvanostatic charge-discharge profiles after the initial and 10000th cycles.

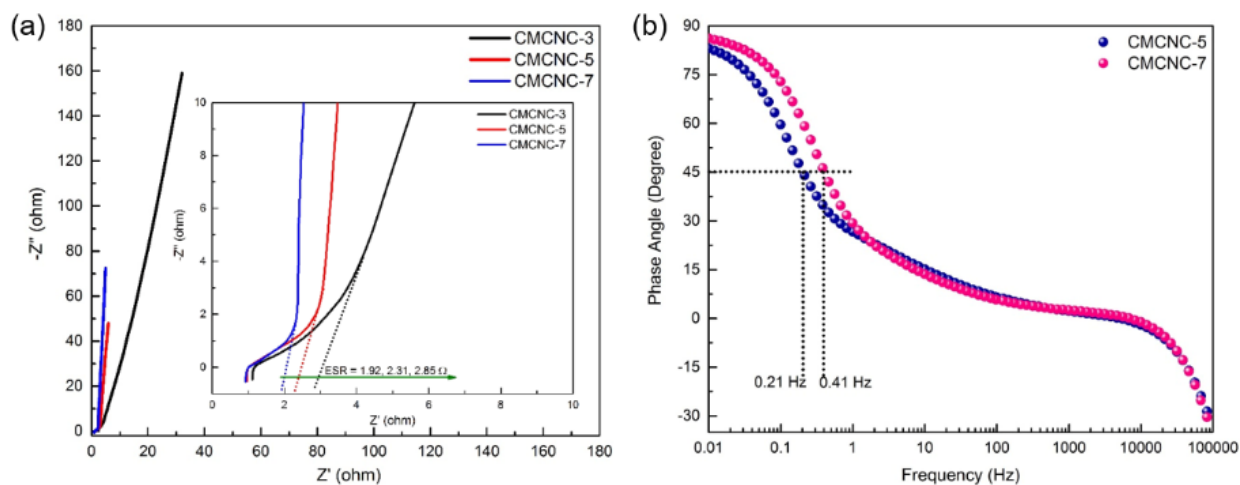


Fig .6. (a) Nyquist plots of the CMCNCs (Inset shows the magnified high-frequency region), (b) frequency response of the phase angle.

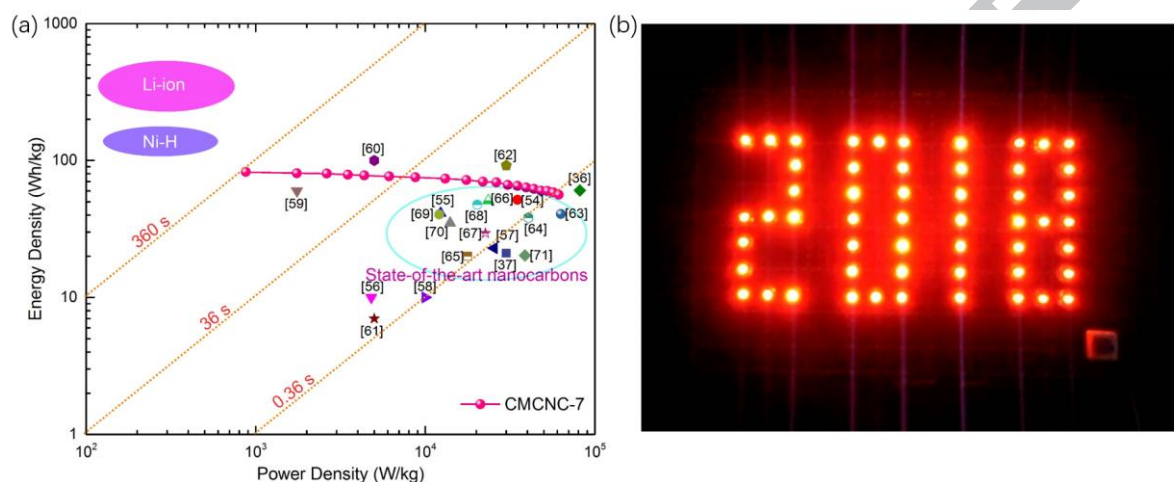


Fig. 7. (a) Ragone plots of CMCNC-7 based symmetric supercapacitor device in comparison with other previously reported nanocarbons based supercapacitors (maximum value) in aqueous or non-aqueous electrolyte, (b) digital photograph of the red LED module was powered by one CMCNC-7 based supercapacitor after charge at 5 A/g for 67s.

Table 1 The specific surface are, total pore volume, mesopore pore volume, mesopore pore volume fraction, and mesopore surface area of the carbon samples.

Sample	SSA	V_t	V_{meso}	V_{meso}/V_t	Meso-SSA
	(m^2/g)	(cm^3/g)	(cm^3/g)	(%)	(m^2/g)
1	744	1.19	1.12	94.1	690
2	983	1.42	1.31	92.2	901
3	1198	1.85	1.69	91.3	1068
4	433	0.56	0.52	92.8	387
AC	1822	0.86	0.08	9.3	123

SSA: specific surface area; V_t : total pore volume; V_{meso} : mesopore pore volume;

Meso-SSA: mesopore surface area.

1. Unusual carbon nanomesh made of interconnected carbon nanocages was prepared.
2. Carbon nanocages have ultrathin partial graphitic shells, of which most have a few layers of graphene.
3. The as-obtained carbon nanomesh with well-developed mesoporous structure has a large specific surface area.
4. The carbon nanomesh made of carbon nanocages can deliver both high energy and high power density in IL electrolyte.

